Deliverable 3

Generative Artificial Intelligence (580694), Spring 2026

Due: October 31, 2026, 23:59
Weight: 30% of the semester project

1. Objective

Deliverable 3 deepens and improves that system by rebuilding it around AI agents. Your team splits the solution into agents with distinct roles, connects them, and shows which prompting techniques inside each agent account for the improvement.

For this deliverable, an agent is a model call with an assigned role, its own prompt, and a defined input and output, which decides what happens next based on what it observes: calling a tool, handing work to another agent, revising its own output, or stopping. A fixed sequence of prompts with no decision inside it is a pipeline and does not qualify as an agent system.

2. Requirements

Agent architecture. Encapsulate the Deliverable 2 solution as a system of at least three agents with distinct roles. For each agent, state its role, the prompt it receives, the tools it can call, the format of what it receives and returns, and the condition under which it stops or hands off. Agents may share the same underlying model.

Model constraint. Every agent in the running system uses the same open-weight model of at most 8 billion parameters.

Prompting techniques. Identify the prompting techniques each agent relies on against a schema. Report results for the direct prompting baseline, the Deliverable 2 solution, the full agent system, and the agent system with each key technique removed one at a time, all on the same set of inputs and under the same correctness criterion.

Execution traces. The system logs the messages exchanged between agents for every input. The traces for the inputs shown in the video and for the failure case are stored in the repository.

3. Format and submission

Video, maximum three minutes. A screen recording of the agent system running on real inputs and showing the pipeline, where the intermediate messages between agents are visible along with the final output. Submit a link with open access.

Technical document. A vertical, single-page PDF compiled from LaTeX. Multi-page submissions and PDFs not produced in LaTeX will be rejected.

Repository. The link works, the code runs from the repository, and the README explains how to reproduce what the video shows. The prompt of each agent is stored as a separate, versioned

file, outside the code.

4. Evaluation rubric

Continuity with Deliverables 1 and 2 (10 points). The agent system attacks the same task and the same failure, under the same definition of a correct output, and builds on the Deliverable 2 solution. Deviations are declared and argued.

Agent graph (10 points). The one-page document includes a graph of how the agents connect. Every agent and every tool appears as a node, every handoff appears as an edge labeled with what is passed, and loops and stopping conditions are marked.

Agent design (10 points). Each agent has a role the task requires, and the other agents do not cover, and the team explains why the work is divided this way.

Prompting techniques (10 points). The techniques used in each agent are named and linked to the diagnosed failure, and their contribution is measured by removing them one at a time.

Evidence of improvement (10 points). Results are reported over a stated set of inputs, applying the same correctness criterion to the baseline, to the Deliverable 2 solution, and to the agent system. One favorable example is an illustration, not evidence.

Repository and reproducibility (10 points). The link works at the time of grading, the code runs from the repository, the prompts of every agent are in versioned files, and the README allows a reader to reproduce what the video shows, traces included.